

Name: Solar Panel Team**Date:** 10/3/2025**Class:** Design for Engineers e4usa

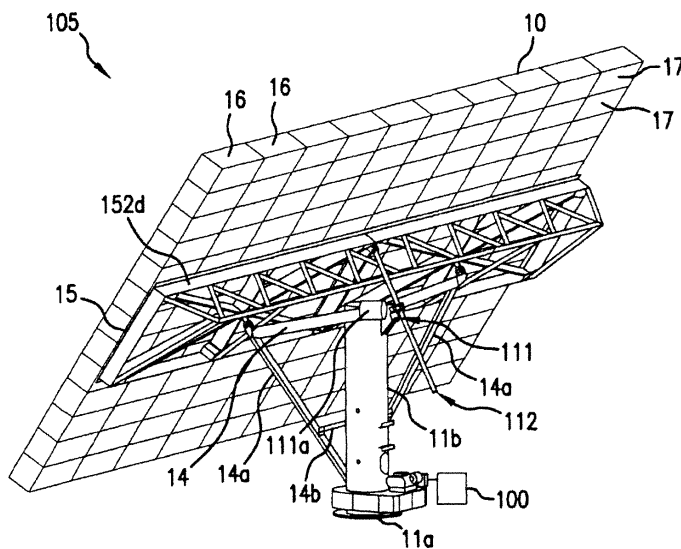
MyDesign Portfolio

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - Solar tracking system

Source: <https://patents.google.com/patent/US8946608B2/en>

(2023). US8946608B2 - solar tracking system. Google Patents.
<https://patents.google.com/patent/US8946608B2/en>



Details

This solar panel tracking system allows solar panels to move with the sun as it rises and sets in order to obtain the maximum amount of sunlight. The movement in the design allows the solar panel to be at the optimal angle at all times of the day, which leads to maximum output.

Analysis

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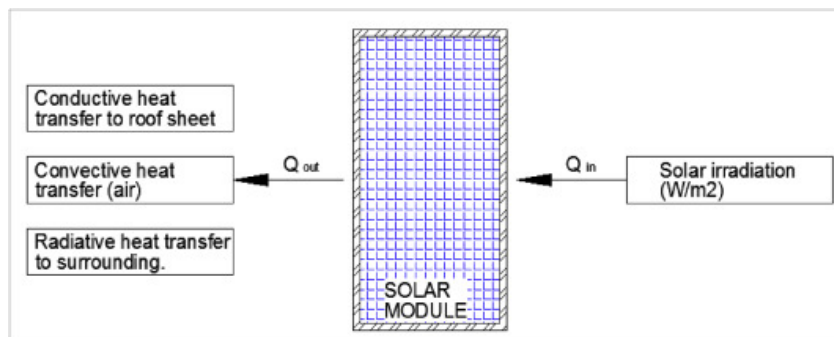
While this design attempt successfully allows solar panels to obtain more energy, it retains the current solar panel structure, which can lead to inefficiencies. Design changes, such as dynamic panel structure, would be optimal in enhancing the solar panels' output, as current panel technology remains limited in maximizing solar panel energy levels.

Prior Design Attempt #2 - Solar Panel Cooling System

Source: <https://www.sciencedirect.com/science/article/pii/S2214157X19305416>

Dwivendi, Pushpendu, et al. "Advanced cooling techniques of P.V. modules: A state of art." *ScienceDirect*, Oct. 2020, www.sciencedirect.com/science/article/pii/S2214157X19305416. Accessed 6 Oct. 2025.

Image



Details

This design attempt looks at different ways to cool solar panels to prevent excessive heat buildup. These include air-based cooling, liquid-based cooling, water spraying, heat pipes, and extended surfaces. Incorporating these into solar panels allows them to cool properly without degrading their performance.

Analysis

All of the concepts provided in the study showed potential in cooling solar panels and increasing the efficiency of solar panels. As a result, they can be integrated into solar panels to improve their performance and prevent it from decreasing over time.

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Prior Design Attempt #3 - Mid-Generation Solar Panels -1980 to 2000-



Gridserve. "6 Reasons Why Solar Energy Is Awesome." GRIDSERVE, 5 Aug. 2022, www.gridserve.com/6-reasons-why-solar-energy-is-awesome/.

Details

The Mid-Generation Solar Panels were made between 1980 and 2000. Due to the commercialization of silicon-based photovoltaic (PV) modules, Mid-Generation solar panels were widespread. These solar panels saw breakthroughs in efficiency, rising 15-20% by the late 1990s. Lastly, these solar panels incorporated thin-film technology, allowing for lightweight and flexible panels.

Analysis

While efficiency greatly improved between 1980 and 2000, it was nowhere near where it is today. In addition, citizens were unwilling to give up space on their land to put solar panels. Lastly, there were environmental concerns with the use of CdTe technology.

Prior Design Attempt #4 - Bifacial Solar Panels

Source

U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy. "Bifacial Solar Panels: Capturing Light from Both Sides."

Longi Solar Technical Paper, Performance Analysis of Bifacial PV Modules under Variable Albedo Conditions.



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PV Magazine, "Bifacial Modules: The Challenges and Advantages"

Details

Bifacial solar panels generate power from both the front and back surfaces. They capture reflected sunlight, called albedo, from the ground or nearby surfaces. They use transparent sheets on the back or glass on both sides to allow more light to reach the cells. Studies have shown that bifacial panels can produce 10 to 20 percent more energy than traditional panels, especially in areas with high albedo.

Analysis

This design improves energy yields without increasing the size of the panel, making it ideal for large installations. Despite this, efficiency usually depends heavily on how reflective the solar panel is and the installation angle. The panels are more expensive due to the



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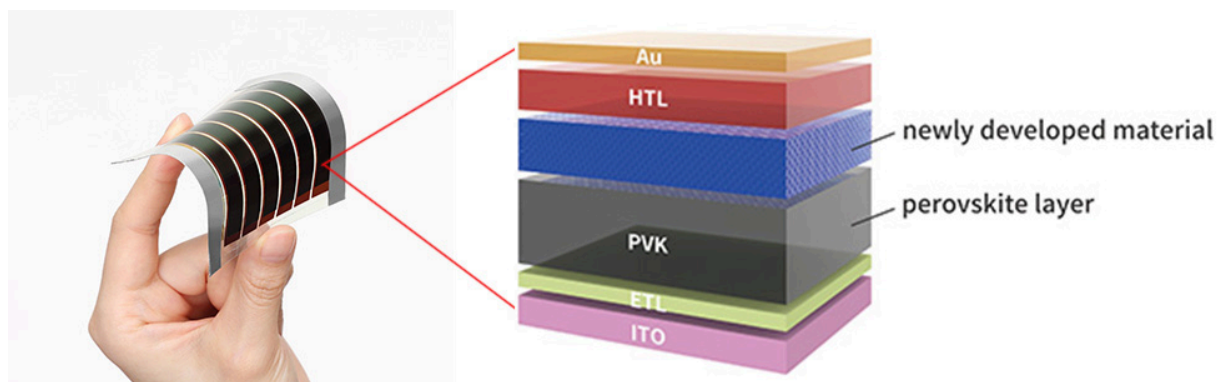
manufacturing challenges of producing both sides of the panel. Future research aims to optimize the efficiency of the back side of the panels and develop cheaper transparent materials to make production less expensive.

Prior Design Attempt #5 - Perovskite Solar Cells

Source

National Renewable Energy Laboratory (NREL). "Perovskite Solar Cells."

Oxford PV. "Tandem Perovskite-Silicon Solar Cell Efficiency Reaches 28%."



Global.Canon: "Canon develops high-performance materials for perovskite solar cells to improve substantial durability and mass-production stability."

Details

Perovskite solar cells use a synthetic crystalline structure that can be altered to absorb different wavelengths of light. They are cheaper to manufacture than silicon solar panels and have achieved higher records of efficiency, especially when layered on top of silicon in "tandem" cells. These efficiencies in the tandem cells have reached 28%. They are also flexible and lightweight, which makes them ideal for new applications like window coatings or portable electronic devices.

Analysis

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While perovskites promise major efficiency gains, their durability remains a key challenge. Moisture, heat, and UV radiation exposure degrade the material quickly, which limits its large-scale deployment. Current research efforts focus on methods to combine silicon and perovskite panel materials that increase durability while maintaining high performance.